

CS2 Analytics Engine

Raw game data in. A coached, actionable performance report out. Fully automated.

The Problem

Every competitive match produces a replay file packed with the data needed to improve, but it is locked in a dense binary format spread across thousands of events. There is no automated path from raw file to specific, actionable feedback. This is the universal operations problem: the answer is in the data, but nothing turns the data into a decision.

The Approach — A 5-Stage Automated Pipeline



CS2 Analytics Engine · fully automated, single command

The Results

2,583
lines of Python

5
analytical dimensions

292
rounds per run

0
manual data entries

Generates a real report with prioritized, data-tied recommendations, plus a touch-friendly web dashboard served by a FastAPI daemon.

What This Demonstrates

- **Data pipeline engineering** — raw binary to parsed events to computed metrics to report
- **Turning data into decisions** — output is prioritized recommendations, not just numbers
- **Sourced, trustworthy output** — every metric traces to raw data, zero manual entry
- **Scope discipline** — built the high-leverage slice, deferred the rest deliberately
- **Full-stack delivery** — Python analysis, FastAPI service, web dashboard

The same pattern I apply to operations: take raw, messy data and build the system that turns it into a decision someone can act on.

Python · demoparser2 · FastAPI · JavaScript dashboard
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